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## INSULATED SLEEVE CONNECTORS AND METHODS FOR CONNECTING DUCTS


### Abstract

An insulated sleeve connector includes a tubular body having a longitudinal axis. The tubular body includes a first end including a first opening, a second end opposite the first end along the longitudinal axis and including a second opening, and a central section situated along the longitudinal axis between the first end and the second end and including an insulating material. The insulated sleeve connector also includes a first protrusion projecting from an inner surface of the tubular body adjacent to the central section and a second protrusion projecting from the inner surface of the tubular body adjacent to the central section and spaced away from the first protrusion along the longitudinal axis.



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## Background/Summary

FIELD

[0001] The present disclosure relates generally to duct systems and, more particularly, to insulated sleeve connectors that connect ducts together and methods for connecting ducts together using insulated sleeve connectors.

BACKGROUND

[0002] Duct systems include ducts that are connected together to form passageways for moving fluid. Often, the ducts are connected together in an end-to-end fashion using a connector and/or clamps. One issue with existing duct systems is that it can be difficult to properly connect the ducts using conventional connectors. Improper connections can result in fluid leaks or disconnection of the ducts. Another issue with existing duct systems is that fluid passing between ends of the connected ducts is subject to ambient temperature. Accordingly, those skilled in the art continue with research and development efforts in duct systems.

SUMMARY

[0003] Disclosed are examples of an insulated sleeve connector for connecting ducts together, a duct system connected using an insulated sleeve connector, and a method for connecting ducts together. The following is a non-exhaustive list of examples, which may or may not be claimed, of the subject matter according to the present disclosure.

[0004] In an example, the disclosed insulated sleeve connector includes a tubular body having a longitudinal axis. The tubular body includes a first end including a first opening, a second end opposite the first end along the longitudinal axis and including a second opening, and a central section situated along the longitudinal axis between the first end and the second end and including an insulating material. The insulated sleeve connector also includes a first protrusion projecting from an inner surface of the tubular body adjacent to the central section and a second protrusion projecting from the inner surface of the tubular body adjacent to the central section and spaced away from the first protrusion along the longitudinal axis.

[0005] In another example, the disclosed insulated sleeve connector includes a tubular body having a longitudinal axis. The tubular body includes a first end including a first opening, a second end opposite the first end along the longitudinal axis and including a second opening, and a central section situated along the longitudinal axis between the first end and the second end and including an insulating material. The insulated sleeve connector also includes a hinge disposed on the tubular body between the central section and the second end. A portion of the tubular body is configured to fold back along the longitudinal axis about the hinge.

[0006] In an example, the duct system includes a first duct, a second duct, and an insulated sleeve connector connecting the first duct and the second duct together. The insulated sleeve connector includes a tubular body having a longitudinal axis. The tubular body includes a first end including a first opening, a second end opposite the first end along the longitudinal axis and including a second opening, and a central section situated along the longitudinal axis between the first end and the second end and including an insulating material. The insulated sleeve connector also includes a first protrusion projecting from an inner surface of the tubular body adjacent to the central section and a second protrusion projecting from the inner surface of the tubular body adjacent to the central section and spaced away from the first protrusion along the longitudinal axis.

[0007] In another example, the duct system includes a first duct, a second duct, and an insulated sleeve connector connecting the first duct and the second duct together. The insulated sleeve connector includes a tubular body having a longitudinal axis. The tubular body includes a first end including a first opening, a second end opposite the first end along the longitudinal axis and including a second opening, and a central section situated along the longitudinal axis between the first end and the second end and including an insulating material. The insulated sleeve connector also includes a hinge disposed on the tubular body between the central section and the second end. A portion of the tubular body is configured to fold back along the longitudinal axis about the hinge.

[0008] In an example, the disclosed method includes steps of: (1) inserting a first duct into a first end of a tubular body of an insulated sleeve connector; (2) moving the first duct end past a first protrusion projecting from an inner surface of the tubular body adjacent to a central section of the tubular body situated along a longitudinal axis of the tubular body between the first end and a second end of the tubular body; (3) positioning a first duct end of the first duct within the central section of the tubular body; (4) inserting a second duct into a second end of the tubular body; (5) moving a second duct end past a second protrusion projecting from the inner surface of the tubular body adjacent to the central section and spaced away from the first protrusion along the longitudinal axis; (6) positioning the second duct end of the second duct within the central section of the tubular body such that a distance between the first duct end and the second duct end is less than a length dimension of the central section along the longitudinal axis; and (7) insulating a gap between the first duct end and the second duct end with the central section.

[0009] In another example, the disclosed method includes steps of: (1) inserting a first duct into a first end of a tubular body of the insulated sleeve connector; (2) positioning a first duct end of the first duct within a central section of the tubular body situated along a longitudinal axis of the tubular body between the first end and a second end of the tubular body; (3) folding a second section of the tubular body, extending along the longitudinal axis between the second end and the central section, back along the longitudinal axis about a hinge; (4) inserting a second duct into the second section of the tubular body; (5) unfolding the second section of the tubular body along the longitudinal axis about the hinge; (6) positioning a second duct end of the second duct within the central section of the tubular body such that a distance between the first duct end and the second duct end is less than a length dimension of the central section along the longitudinal axis; and (7) insulating a gap between the first duct end and the second duct end with the central section.

[0010] Also disclosed are examples of an aircraft that includes a duct system connected using an insulated sleeve connector.

[0011] Other examples of the insulated sleeve connector, the duct system, and the method will become apparent from the following detailed description, the accompanying drawings, and the appended claims.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. **1** is a schematic block diagram of an example of an insulated sleeve connector;

[0013] FIG. **2** is a flow diagram of an example of a method for connecting ducts;

[0014] FIG. **3** is a schematic, perspective view of an example of the insulated sleeve connector;

[0015] FIG. **4** is a schematic, perspective view of an example of the insulated sleeve connector connecting a first duct and a second duct together;

[0016] FIG. **5** is a schematic, section view of an example of the insulated sleeve connector;

[0017] FIG. **6** is a schematic, section view of an example of the insulated sleeve connector connecting a first duct and a second duct together;

[0018] FIG. **7** is a schematic, section view of an example of the insulated sleeve connector connected to a first duct;

[0019] FIG. **8** is a schematic, section view of the example of the insulated sleeve connector shown in FIG. **7** connecting the first duct and a second duct together;

[0020] FIG. **9** is a schematic, section view of an example of the insulated sleeve connector;

[0021] FIG. **10** is a schematic, section view of an example of the insulated sleeve connector

connecting a first duct and a second duct together;
[0022] FIG. **11** is a schematic, perspective view of an example of the insulated sleeve connector;
[0023] FIG. **12** is a schematic illustration of an example of a portion of a reinforcing band of the insulated sleeve connector shown in FIG. **11**;
[0024] FIG. **13** is a schematic illustration of an example of the portion of the reinforcing band of the insulated sleeve connector shown in FIG. **12** in an expanded state;
[0025] FIG. **14** is a schematic, section view of an example of the insulated sleeve connector connecting a first duct and a second duct together;
[0026] FIG. **15** is a schematic, section view of an example of the insulated sleeve connector connecting a first duct and a second duct together;
[0027] FIG. **16** is a flow diagram of an example of an aircraft manufacturing and service method; and
[0028] FIG. **17** is a schematic illustration of an example of an aircraft.

DETAILED DESCRIPTION

[0029] Referring generally to FIGS. **1-15**, by way of examples, the present disclosure is directed to an insulated sleeve connector **100**, a duct system **250** that includes ducts **252** connected using the insulated sleeve connector **100**, and a method **1000** for connecting the ducts **252** using the insulated sleeve connector **100**. Examples of the insulated sleeve connector **100** and method **1000** provide efficient connection of the ducts **252** to prevent fluid from escaping the duct system **250** and effective insulation of the fluid passing between the ducts **252** connected using the insulated sleeve connector **100**.
[0030] Generally, the duct system **250** includes a first duct **254** and a second duct **256** (e.g., ducts **252**), which are connected in an end-to-end orientation using examples the of insulated sleeve connector **100**. In one particular example, the duct system **250** is an environmental control system (ECS) duct system on an aircraft. Examples of the insulated sleeve connector **100** connect the ECS ducts (e.g., ducts **252**) in the end-to-end orientation. In various examples, the insulated sleeve connector **100** has an opening on each end and a number of pull tabs located around the perimeter of each opening. In various examples, the opening at each end of the insulated sleeve connector **100** is chamfered to facilitate installation onto the duct. In various examples, a middle section of the insulated sleeve connector **100** is suitably sized to ensure minimum duct separation requirements are met. In various examples, the middle section of the insulated sleeve connector **100** has a thickness and/or is made of a material suitable to insulate the entire duct joint.
[0031] Referring to FIG. **1**, in one or more examples, the insulated sleeve connector **100** includes a number of elements, features, and components. The insulated sleeve connector **100** includes a tubular body **102**. In various examples, at least a portion of the tubular body **102** includes or is made of an insulating material **124**. In various examples, the tubular body **102** includes one or more of a first opening **114**, a second opening **118**, a first end **112**, a second end **116**, a first section **126**, a second section **128**, and a central section **122**. In various examples, the insulated sleeve connector **100** also includes one or more of a first protrusion **160**, a second protrusion **170**, a first groove **132**, a second groove **134**, a hinge **164**, a first retainer **166**, a second retainer **168**, a first inner recess **196**, a second inner recess **198**, a first outer recess **174**, a second outer recess **176**, a ramp **172**, a second ramp **204**, a plurality of first tabs **152**, a plurality of second tabs **154**, a first inner band **186**, a second inner band **188**, a first outer band **192**, and a second outer band **194**, among other elements, features, and components.
[0032] Referring to FIGS. **1** and **3-15**, the following are examples of the insulated sleeve connector **100**, according to the present disclosure. Not all of the elements, features, and/or components described or illustrated in one example are required in that example. Some or all of the elements, features, and/or components described or illustrated in one example can be combined with other examples in various ways without the need to include other elements, features, and/or components described in those other examples, even though such combination or combinations are not

explicitly described or illustrated by example herein.

[0033] Referring to FIGS. **1** and **3-15**, in one or more examples, the insulated sleeve connector **100** includes the tubular body **102**. The tubular body **102** has a longitudinal axis **104**. The tubular body **102** includes the first end **112**. The tubular body **102** includes the first opening **114** at the first end **112**. The tubular body **102** includes the second end **116**. The second end **116** is opposite the first end **112** along the longitudinal axis **104**. The tubular body **102** includes the second opening **118** at the second end **116**. The tubular body **102** includes the central section **122**. The central section **122** is situated along the longitudinal axis **104** between the first end **112** and the second end **116**. The central section **122** includes or is made of the insulating material **124**.

[0034] In various examples, the insulated sleeve connector **100** is configured to position the ducts **252** in an end-to-end orientation. In one or more examples, a distance **286** between the ends of the ducts **252** can vary. In one or more examples, the insulated sleeve connector **100** aligns the longitudinal axis of the first duct **254** to be co-axial with the longitudinal axis of the second duct **256**. In other examples, the ducts **252** are aligned in an offset angular orientation (e.g., the longitudinal axes of the first duct **254** and the second duct **256** are not co-axial). In one or more examples, the insulated sleeve connector **100** is configured to space apart the ends of the first duct **254** and the second duct **256** to form a gap **258** between a first duct end **272** and a second duct end **274**.

[0035] In one or more examples, the tubular body **102** of the insulated sleeve connector **100** has a cylindrical shape and forms a hollow or open interior space. In one or more examples, an inner dimension **142** (e.g., inner diameter) of the tubular body **102** is constant along the longitudinal axis **104**. In one or more examples, the inner dimension **142** varies along the longitudinal axis **104**. In one or more examples, the inner dimension **142** of the tubular body **102** at the first end **112** (e.g., the first opening **114**) and at the second end **116** (e.g., the second opening **118**) is the same. In one or more examples, the inner dimension **142** of the tubular body **102** at the first end **112** (e.g., the first opening **114**) and at the second end **116** (e.g., the second opening **118**) is different. Generally, the inner dimension **142** of the tubular body **102** at the first end **112** (e.g., the first opening **114**) and at the second end **116** (e.g., the second opening **118**) is configured to provide a snug fit around the first duct **254** and the second duct **256**. In one or more examples, the insulated sleeve connector **100** is constructed from a material that expands during insertion of the ducts **252**. In various examples, one or more portions or sections of the tubular body **102** can be made of fluorinated ethylene propylene (FEP), polytetrafluoroethylene (PTFE), plasticized vinyl, perfluoroalkoxy (PFA), polyvinylidene fluoride (PVDF), and the like.

[0036] As illustrated in FIGS. **4**, **6-8**, **9**, **14** and **15**, the insulated sleeve connector **100** connects the first duct **254** and the second duct **256** together and prevents fluid from escaping between the ducts **252**. The tubular body **102** is an elongated body that includes or forms a hollow interior space configured to receive the ducts **252**.

[0037] The first opening **114** of the tubular body **102** is configured to receive the first duct end **272** of the first duct **254** such that at least a first portion **282** of the first duct **254** is positioned within the first section **126** of the tubular body **102**. The second opening **118** of the tubular body **102** is configured to receive a second duct end **274** of the second duct **256** such that at least a second portion **284** of the second duct **256** is positioned within the second section **128** of the tubular body **102**.

[0038] In one or more examples, the tubular body **102** has a length, measured along the longitudinal axis **104**, such that at least a portion of the first section **126** overlaps the first portion **282** of the first duct **254** and at least a portion of the second section **128** overlaps the second portion **284** of the second duct **256** and the central section **122** spans the gap **258** between the first duct end **272** and the second duct end **274**.

[0039] In various examples, the insulated sleeve connector **100** properly positions the first duct end **272** and the second duct end **274** relative to each other when connecting the first duct **254** and the second duct **256** together. In various examples, the central section **122** insulates the gap **258** between the first duct end **272** and the second duct end **274**. In various examples, the first section **126** and/or the second section **128** also insulate end portions of the ducts **252**. The insulating effect is achieved by selection of the insulating material **124** used to form at least a portion of the insulated sleeve connector **100**, a thickness of the tubular body **102**, a thickness of the insulating material **124** used, and the like.

[0040] Referring generally to FIG. **1** and particularly to FIGS. **5**, **6**, **9** and **10**, in one or more examples, the insulated sleeve connector **100** includes the first protrusion **160**. The first protrusion **160** projects (e.g., radially inward) from the inner surface **136** of the tubular body **102**. The first protrusion **160** is adjacent to the central section **122**. The insulated sleeve connector **100** includes the second protrusion **170**. The second protrusion **170** projects (e.g., radially inward) from the inner surface **136** of the tubular body **102**. The second protrusion **170** is adjacent to the central section **122** and is spaced away from the first protrusion **160** along the longitudinal axis **104**.

[0041] The first protrusion **160** serves to retain the first duct **254** within the first section **126** of the tubular body **102**. Additionally, or alternatively, the first protrusion **160** also provides a tactile response (e.g., a snap) that indicates appropriate (e.g., full) insertion of the first duct **254** within the tubular body **102** and, thus, proper installation and connection of the insulated sleeve connector **100**. Similarly, the second protrusion **170** serves to retain the second duct **256** within the second section **128** of the tubular body **102**. Additionally, or alternatively, the second protrusion **170** also provides tactile response (e.g., a snap) that indicates appropriate (e.g., full) insertion of the second duct **256** within the tubular body **102** and, thus, proper installation and connection of the insulated sleeve connector **100**.

[0042] In one or more examples, the first protrusion **160** and/or the second protrusion **170** have any suitable shape or structural configuration. As examples, the first protrusion **160** and/or the second protrusion **170** include or take the form of a bump, a tooth, a finger, and the like. In one or more examples, the first protrusion **160** and/or the second protrusion **170** are annular and extend continuously along a perimeter of the inner surface **136**. In one or more examples, the first protrusion **160** and/or the second protrusion **170** are discontinuous and are formed by a series of protrusions that extend along a perimeter of the inner surface **136**. In one or more examples, the first protrusion **160** and/or the second protrusion **170** includes a number of adjacent protrusions that extend along the longitudinal axis **104**. In one or more examples, the first protrusion **160** and/or the second protrusion **170** are formed by or represent a reduction in an inner dimension **142** (e.g., inner diameter) of the tubular body **102** relative to other portions of the tubular body **102**.

[0043] Referring generally to FIG. **1** and particularly to FIGS. **7** and **8**, in one or more examples, the insulated sleeve connector **100** includes the hinge **164**. The hinge **164** is disposed on the tubular body **102** between the central section **122** and the second end **116**. A portion of the tubular body **102** is configured to fold back upon itself along the longitudinal axis **104** about the hinge **164** and then unfold to its original configuration.

[0044] FIG. **7** illustrates an example of the insulated sleeve connector **100** with a portion of the tubular body **102** folded back upon itself along the hinge **164**. FIG. **8** illustrates an example of the insulated sleeve connector **100** shown in FIG. **7** connected to the second duct **256**. The hinge **164** enables length of the tubular body **102** along the longitudinal axis **104** to be temporarily reduced for installation of the insulated sleeve connector **100**, such as during insertion of one of the ducts **252** when connecting the ducts **252** together using the insulated sleeve connector **100**. The hinge **164** can include or take the form of any structure capable of enabling one portion of the tubular body **102** to fold or roll back upon itself and then return to its original position. In one or more examples, the hinge **164** is annular and extends continuously along a perimeter of the inner surface **136** or an outer surface **138** of the tubular body **102**. Examples of the hinge **164** include a recess formed in the inner surface **136** of the tubular body **102** (e.g., a reduction in the inner dimension **142**), a recess formed in the outer surface **138** of the tubular body **102** (e.g., a reduction in an outer dimension **144**), a score formed along a perimeter of the inner surface **136** and/or the outer surface **138**, a living hinge, and the like.

[0045] Referring to FIG. **1**, in one or more examples of the insulated sleeve connector **100**, the insulating material **124** includes or is made of foam **156**. In one or more examples of the insulated sleeve connector **100**, the insulating material **124** includes or is made of closed cell foam **158**. In one or more examples, the insulating material **124** includes or is made of silicone. In one or more examples, the insulating material **124** includes or is made of polyvinylidene fluoride or polyvinylidene difluoride (PVDF). In one or more examples, the insulating material **124** includes or is made of polyethylene (PE) or the like. In one or more examples of the insulated sleeve connector **100**, at least one of the first section **126** and the second section **128** of the tubular body **102** includes or is made of the insulating material **124**.

[0046] Referring to FIGS. **1**, **3-11**, **14** and **15**, in one or more examples of the insulated sleeve connector **100**, the tubular body **102** includes the first section **126**. The first section **126** extends along the longitudinal axis **104** between the first end **112** and the central section **122**. The tubular body **102** includes the second section **128**. The second section **128** extends along the longitudinal axis **104** between the second end **116** and the central section **122**. The first section **126** of the tubular body **102** is configured to receive the first duct **254**. The second section **128** of the tubular body **102** is configured to receive the second duct **256**.

[0047] Referring generally to FIG. **1** and particularly to FIG. **6**, in one or more examples of the insulated sleeve connector **100**, the central section **122** of the tubular body **102** has a length dimension **162**. The length dimension **162** is measured along the longitudinal axis **104**. The length dimension **162** is greater than a distance **286** between the first duct end **272** of the first duct **254** and the second duct end **274** of the second duct **256** when the first duct **254** is received by the first section **126** and the second duct **256** is received by the second section **128**. The distance **286** is measured along the longitudinal axis **104** of the tubular body **102**.

[0048] The length dimension **162** of the central section **122** of the tubular body **102** being greater than the distance **286** between the first duct end **272** of the first duct **254** and the second duct end **274** of the second duct **256** insulates the gap **258** between the first duct end **272** of the and the second duct end **274** and, thus, insulates the fluid passing between the first duct **254** and the second duct **256**.

[0049] Referring to FIGS. **1**, **5**, **6**, **9** and **10**, in one or more examples of the insulated sleeve connector **100**, the first protrusion **160** is located at a first junction of the first section **126** and the central section **122**. The first protrusion **160** provides a first tactile response to insertion of the first duct **254** and assists in retaining the first duct **254** within the first section **126**. The second protrusion **170** is located at a second junction of the second section **128** and the central section **122**. The second protrusion **170** provides a second tactile response to insertion of the second duct **256** and assists in retaining the second duct **256** within the second section **128**. Additionally, in one or more examples, the tactile response (e.g., a snap) provided by the first protrusion **160** and/or the second protrusion **170** also indicates appropriate (e.g., full and correct) insertion of the first duct **254** and/or the second duct **256** within the tubular body **102** and, thus, proper installation and connection of the insulated sleeve connector **100** on the ducts **252**.

[0050] As illustrated in FIGS. **5** and **6**, in various examples of the duct system **250**, the first duct **254** includes a first bead **262** that extends or projects radially outward from an outer surface of the first duct **254**. Similarly, in various examples of the duct system **250**, the second duct **256** includes a second bead **264** that extends or projects radially outward from an outer surface of the second duct **256**. In some examples, the first bead **262** is located at the first duct end **272** of the first duct **254** and/or the second bead **264** is located at the second duct end **274** of the second duct **256**. In other examples, the first bead **262** is spaced inward of the first duct end **272** of the first duct **254** and/or the second bead **264** is spaced inward of the second duct end **274** of the second duct **256**. The first bead **262** and the second bead **264** can have various structural configurations and can include, for example, beads, bends, or other structures formed by the ducts **252** or connected to the ducts **252**. In the illustrated examples, the first bead **262** is a bead formed along a perimeter of the outer surface of the first duct **254** and the second bead **264** is a bead formed along a perimeter of the outer surface of the second duct **256**. In one or more examples, the tubular body **102** of the insulated sleeve connector **100** is configured to accommodate the first bead **262** and the second bead **264** during insertion of the ducts **252** and installation of the insulated sleeve connector **100**.

[0051] Referring to FIGS. **1**, **5** and **6**, in one or more examples, the insulated sleeve connector **100** includes the first groove **132**. The first groove **132** is disposed on the inner surface **136** of the tubular body **102** along the central section **122**, wherein the first groove **132** is configured to receive the first bead **262** of the first duct **254**. The insulated sleeve connector **100** also includes the second groove **134**. The second groove **134** is disposed on the inner surface **136** of the tubular body **102** along the central section **122**. The second groove **134** is spaced from the first groove **132** along the longitudinal axis **104**. The second groove **134** is configured to receive the second bead **264** of the second duct **256**.

[0052] As illustrated in FIGS. **5** and **6**, the first groove **132** is configured to receive and capture the first bead **262** of the first duct **254** as the first duct **254** is moved along the longitudinal axis **104** relative to the tubular body **102**. The first groove **132** is also configured to retain the first bead **262** and, thus, the first duct end **272** of the first duct **254** at an appropriate (e.g., fully inserted) position within the tubular body **102** and relative to the second duct end **274** of the second duct **256** when the ducts **252** are connected together such that a proper instance of the distance **286** of the gap **258** between the ducts **252** is maintained. Similarly, the second groove **134** is configured to receive and capture the second bead **264** of the second duct **256** as the second duct **256** is moved along the longitudinal axis **104** relative to the tubular body **102**. The second groove **134** is also configured to retain the second bead **264** and, thus, the second duct end **274** of the second duct **256** at an appropriate (e.g., fully inserted) position within the tubular body **102** and relative to the first duct end **272** of the first duct **254** when the ducts **252** are connected together such that a proper instance of the distance **286** of the gap **258** between the ducts **252** is maintained.

[0053] In one or more examples, the first groove **132** also provides a tactile response (e.g., a snap) that indicates appropriate (e.g., full) insertion of the first duct **254** within the tubular body **102** and, thus, proper installation and connection of the insulated sleeve connector **100**. Similarly, the second groove **134** provides tactile response (e.g., a snap) that indicates appropriate (e.g., full) insertion of the second duct **256** within the tubular body **102** and, thus, proper installation and connection of the insulated sleeve connector **100**.

[0054] In one or more examples, the first groove **132** depends (e.g., radially inward) from the inner surface **136** of the central section **122** of the tubular body **102**. In one or more examples, the first groove **132** is adjacent to an interface between the central section **122** and the first section **126**. In one or more examples, an outward end of the first groove **132** includes or is formed by the first protrusion **160**. In one or more examples, the second groove **134** depends (e.g., radially inward) from the inner surface **136** of the central section **122** of the tubular body **102**. In one or more examples, the second groove **134** is adjacent to an interface between the central section **122** and the second section **128**. In one or more examples, an outward end of the second groove **134** includes or is formed by the second protrusion **170**.

[0055] In one or more examples, the first groove **132** and/or the second groove **134** have any suitable shape or structural configuration. As examples, the first groove **132** and/or the second groove **134** include or take the form of a recess, a slot, a depression, and the like. In one or more examples, the first groove **132** and/or the second groove **134** are annular and extend continuously along a perimeter of the inner surface **136** of the central section **122**. In one or more examples, the first groove **132** and/or the second groove **134** are discontinuous and are formed by a series of gouges that extend along a perimeter of the inner surface **136**. In one or more examples, the first groove **132** and/or the second groove **134** are formed by or represent a reduction in the inner dimension **142** (e.g., inner diameter) of the central section **122** of tubular body **102** relative to other portions (e.g., outward and/or inward portions) of the central section **122** of the tubular body **102**.

[0056] Referring to FIGS. **9** and **10**, in other examples, the insulated sleeve connector **100** includes a single instance of a groove **288**. The groove **288** is disposed on the inner surface **136** of the tubular body **102** along the central section **122**. In these examples, the groove **288** is configured to receive, capture, and/or retain the first bead **262** of the first duct **254** and the second bead **264** of the second duct **256**. In one or more examples, opposing ends of the groove **288** include or are formed by the first protrusion **160** and the second protrusion **170**. In one or more examples, the groove **288** is formed by or represents a reduction in the inner dimension **142** (e.g., inner diameter) of the central section **122** of tubular body **102** relative to other portions (e.g., outward portions) of the central section **122** of the tubular body **102**.

[0057] Referring to FIGS. **1**, **5** and **6**, in one or more examples of the insulated sleeve connector **100**, the first groove **132** has a first groove width **146** (FIG. **5**). The first groove width **146** measured along the longitudinal axis **104**. The first groove width **146** is larger than a first bead width **266** (FIG. **6**) of the first bead **262**. Similarly, in one or more examples of the insulated sleeve connector **100**, the second groove **134** has a second groove width **148** (FIG. **5**.) The second groove width **148** is measured along the longitudinal axis **104**. The second groove width **148** is larger than a second bead width **268** (FIG. **6**) of the second bead **264**.

[0058] The first groove width **146** being slightly larger than the first bead width **266** enables the first groove **132** to efficiently receive and retain the first bead **262**. The first groove width **146** being larger than the first bead width **266** also permits the first duct **254** to shift or move a small distance relative to the insulated sleeve connector **100**. The second groove width **148** being slightly larger than the second bead width **268** enables the second groove **134** to efficiently receive and retain the second bead **264**. The second groove width **148** being larger than the second bead width **268** also permits the second duct **256** to shift or move a small distance relative to the insulated sleeve connector **100**.

[0059] Referring to FIGS. **1**, **7** and **8**, in one or more examples of the insulated sleeve connector **100**, the second section **128** of the tubular body **102** includes the hinge **164**. As illustrated in FIG. **7**, in these examples, the hinge **164** is located inward of the second end **116**. A portion (e.g., an outward portion) of the second section **128** is configured to fold back upon another portion (e.g., an inward portion) of the second section **128** along the longitudinal axis **104** about the hinge **164**. In the hinged state, or inwardly folded configuration, the length of the tubular body **102** along the longitudinal axis **104** is reduced such that the second duct end **274** of the second duct **256** can be more easily inserted in the second section **128** and positioned relative to the central section **122**. As illustrated in FIG. **8**, after the second duct **256** is installed, the folded portion of the second section **128** can be unfolded or returned to its original configuration.

[0060] Referring to FIGS. **1** and **4**, in one or more examples, the insulated sleeve connector **100** includes at least one of the first retainer **166** and/or the second retainer **168**. The first retainer **166** and/or the second retainer **168** are applied to the tubular body **102** and are configured to apply a radially inward compressive force that secures and maintains the tubular body **102** around the ducts **252**. In one or more examples, the first retainer **166** and the second retainer **168** have an annular shape and extend around a perimeter of the outer surface **138** of the first section **126** and the second section **128**, respectively. The first retainer **166** and/or the second retainer **168** can include any suitable clamp, strap, fitting, O-ring, cable tie, and the like.

[0061] Referring to FIGS. **1**, **7** and **8**, in one or more examples, the insulated sleeve connector **100** includes at least the second retainer **168**. The hinge **164** is configured to receive the second retainer **168**. As an example, the hinge **164** can include or take the form of a recess or channel formed in the outer surface **138** of the second section **128**. The second retainer **168** is configured to be seated or received within the open space formed by the hinge **164**. In one or more of these examples, the second retainer **168** includes or takes the form of an O-ring or other flexible retainer.

[0062] Referring still to FIGS. **1**, **7** and **8**, in one or more examples of the insulated sleeve connector **100**, the tubular body **102** has an outer dimension **144** (FIG. **8**) along the first section **126**. The outer dimension **144** along at least a portion of the first section **126** increases from the first end **112** to the central section **122** to form the ramp **172**. The second retainer **168** is movable along the ramp **172** between the first section **126** and the second section **128**.

[0063] Referring still to FIGS. **1**, **7** and **8**, in one or more examples of the insulated sleeve connector **100**, the outer dimension **144** of a portion of the tubular body **102** along at least a portion of the second section **128** increases from the second end **116** to the central section **122** to form a second ramp **204**. As an example, the outer dimension **144** along a portion of the second section **128** increases from an inward end of the hinge **164** to the central section **122** to form the second ramp **204** (e.g., as shown in FIG. **8**). In these examples, the ramp **172** can also be referred to as a first ramp. The second retainer **168** is movable along the second ramp **204** between the second section **128** and the first section **126**.

[0064] Movement of the second retainer **168** from the second section **128** (e.g., as shown in FIG. **8**) to the first section **126** (e.g., as shown in FIG. **7**), for example, over the second ramp **204**, the central section **122**, and the ramp **172**, repositions the second retainer **168** (e.g., O-ring) such that the second section **128** can be inwardly folded about the hinge **164**. Movement of the second retainer **168** from the first section **126** (e.g., as shown in FIG. **7**) back to the second section **128** (e.g., as shown in FIG. **8**), for example, over the ramp **172**, the central section **122**, and the second ramp **204**, repositions the second retainer **168** (e.g., O-ring) back into the hinge **164** to maintain the second section **128** around the second duct **256**.

[0065] Referring to FIGS. **1**, **4-6**, **9-11**, **14** and **15**, in one or more examples, the insulated sleeve connector **100** includes a number of first tabs **152**. The first tabs **152** extend outwardly from and beyond the first end **112** of the tubular body **102** along the longitudinal axis **104**. The first tabs **152** serve as grips that provide a structure for a user to grasp and apply a force to the tubular body **102**, for example, to slide the first section **126** onto the first duct end **272** of the first duct **254**, when installing the insulated sleeve connector **100** on the first duct **254**.

[0066] Referring still to FIGS. **1**, **4-6**, **9-11** and **15**, in one or more examples, the insulated sleeve connector **100** includes a number of second tabs **154**. The second tabs **154** extend outwardly from and beyond the second end **116** of the tubular body **102** along the longitudinal axis **104**. The second tabs **154** serve as grips that provide a structure for a user to grasp, for example, to slide the first section **126** onto the first duct end **272** of the first duct **254**, when installing the insulated sleeve connector **100** on the second duct **256**.

[0067] Referring to FIGS. **4-6**, **9-11** and **15**, in one or more examples of the insulated sleeve connector **100**, in one or more examples, the first tabs **152** extend from a perimeter of the first opening **114**. In one or more examples, the first tabs **152** are spaced apart from each other around the perimeter of the first opening **114**. In one or more examples, each one of the first tabs **152** is equally angularly spaced from a directly adjacent one of the first tabs **152** around the first opening **114**.

[0068] Referring to FIGS. **4-6**, **9-11** and **15**, in one or more examples of the insulated sleeve connector **100**, the second tabs **154** extend from a perimeter of the second opening **118**. In one or more examples, the second tabs **154** are spaced apart from each other around the perimeter of the second opening **118**. In one or more examples, each one of the second tabs **154** is equally angularly spaced from a directly adjacent one of the second tabs **154** around the second opening **118**.

[0069] Referring to FIGS. **4-6**, **9-11** and **15**, in one or more examples of the insulated sleeve connector **100**, at least a portion of at least one of the first tabs **152** is oriented at an oblique angle relative to the longitudinal axis **104**. For example, the first tabs **152** are flared radially outward. In one or more examples, the first tabs **152** are aligned at least approximately parallel to the longitudinal axis **104**.

[0070] Referring to FIGS. **4-6**, **9-11** and **15**, in one or more examples of the insulated sleeve connector **100**, at least a portion of at least one of the second tabs **154** is oriented at an oblique angle relative to the longitudinal axis **104**. For example, the second tabs **154** are flared radially outward. In one or more examples, the second tabs **154** are aligned at least approximately parallel to the longitudinal axis **104**.

[0071] Referring to FIGS. **3-6**, in one or more examples, tubular body **102** includes a first flared portion at the first end **112** and a second portion section at the second end **116**. In these examples, the flared portions extend outward along the longitudinal axis **104** away from the first section **126** and the second section **128**, respectively. The first flared portion represents an increase in the inner dimension **142** (e.g., inner diameter) of the outward portion of the first section **126** relative to the inward portion of the first section **126**. The second flared portion represents an increase in the inner dimension **142** (e.g., inner diameter) of the outward portion of the second section **128** relative to the inward portion of the second section **128**. The flared portions serve as a funnel to direct the ends of the ducts **252** and facilitate insertion of the first duct **254** and second duct **256** into the tubular body **102** of the insulated sleeve connector **100**. In some examples, the first flared portion also serves to maintain the first retainer **166** on the first section **126**. In some examples, the second flared portion also serves to maintain the second retainer **168** on the second section **128** (e.g., as shown in FIG. **4**).

[0072] Referring to FIGS. **3-6**, in one or more examples, the first tabs **152** extend outward from the first flared section of the first section **126**. In one or more examples, the second tabs **154** extend outward from the second flared portion of the second section **128**.

[0073] Referring to FIGS. **1** and **9-11**, in one or more examples, the insulated sleeve connector **100** includes the first flange **182**. The first flange **182** extends from the first section **126** of the tubular body **102** at the first end **112**. In one or more examples, the first flange **182** projects radially outward from the outer surface **138** of the first section **126**. In one or more examples, the first flange **182** is located at or near the first end **112**. In one or more examples, the first flange **182** is annular and extends around the perimeter of the outer surface **138** of the first section **126**. In these examples, the first flange **182** serves to maintain the first retainer **166** on the first section **126**.

[0074] Referring to FIGS. **1** and **9-11**, in one or more examples, the insulated sleeve connector **100** includes the second flange **184**. The second flange **184** extends from the second section **128** of the tubular body **102** at the second end **116**. In one or more examples, the second flange **184** projects radially outward from the outer surface **138** of the second section **128**. In one or more examples, the second flange **184** is located at or near the second end **116**. In one or more examples, the second flange **184** is annular and extends around the perimeter of the outer surface **138** of the second section **121286**. In these examples, the second flange **184** serves to maintain the second retainer **168** on the second section **128**.

[0075] Referring to FIGS. **1** and **11**, in one or more examples, the insulated sleeve connector **100** includes the first inner band **186**. The first inner band **186** is coupled to the inner surface **136** of the first section **126** of the tubular body **102**. In one or more examples, the first inner band **186** is located proximate (e.g., at or near) the first end **112**. In these examples, the first inner band **186** protects the material of the tubular body **102** from wear or damage. Additionally, or alternatively, the first inner band **186** structurally reinforces or strengthens a portion of the first section **126**, such as the portion that receives the first retainer **166**.

[0076] Referring to FIGS. **1** and **11**, in one or more examples, the insulated sleeve connector **100** includes the second inner band **188**. The second inner band **188** is coupled to the inner surface **136** of the second section **128** of the tubular body **102**. In one or more examples, the second inner band **188** is located proximate (e.g., at or near) the second end **116**. In these examples, the second inner band **188** protects the material of the tubular body **102** from wear or damage. Additionally, or alternatively, the second inner band **188** structurally reinforces or strengthens a portion of the second section **128**, such as the portion that receives the second retainer **168**.

[0077] Referring to FIGS. **1** and **11**, in one or more examples, the insulated sleeve connector **100** includes the first outer band **192**. The first outer band **192** is coupled to the outer surface **138** of the first section **126** of the tubular body **102**. In one or more examples the first outer band **192** is located proximate (e.g., at or near) the first end **112**. In one or more examples, the first outer band **192** is located opposite the first inner band **186**. In these examples, the first outer band **192** protects the material of the tubular body **102** from wear or damage. Additionally, or alternatively, the first outer band **192** structurally reinforces or strengthens a portion of the first section **126**, such as the portion that receives the first retainer **166**.

[0078] Referring to FIGS. **1** and **11**, in one or more examples, the insulated sleeve connector **100** includes the second outer band **194**. The second outer band **194** is coupled to the outer surface **138** of the second section **128** of the tubular body **102**. In one or more examples, the second outer band **194** is located proximate (e.g., at or near) the second end **116**. In one or more examples, the second outer band **194** is located opposite the second inner band **188**. In these examples, the second outer band **194** protects the material of the tubular body **102** from wear or damage. Additionally, or alternatively, the second outer band **194** structurally reinforces or strengthens a portion of the second section **128**, such as the portion that receives the second retainer **168**.

[0079] Referring to FIGS. **1** and **11**, in one or more examples, the insulated sleeve connector **100** includes the first outer recess **174**. The first outer recess **174** is disposed on (e.g., depends radially inward from) the outer surface **138** of the first section **126** of the tubular body **102**, for example, proximate (e.g., at or near) the first end **112**. In one or more examples of the insulated sleeve connector **100**, the first outer band **192** is disposed in the first outer recess **174**. In one or more examples, the first retainer **166** is also disposed in the first outer recess **174** over the first outer band **192**. In these examples, the first outer recess **174** serves to retain the first retainer **166** around the first section **126**. In some examples, the first outer recess **174** represents a reduction in the outer dimension **144** (e.g., outer diameter) of a portion of the first section **126** relative to another portion of the first section **126**.

[0080] Referring to FIGS. **1** and **11**, in one or more examples, the insulated sleeve connector **100** includes the second outer recess **176**. The second outer recess **176** is disposed on (e.g., depends radially inward from) the outer surface **138** of the second section **128** of the tubular body **102**, for example, proximate (e.g., at or near) the second end **116**. In one or more examples of the insulated sleeve connector **100**, the second outer band **194** is disposed in the second outer recess **176**. In one or more examples, the second retainer **168** is also disposed in the second outer recess **176** over the second outer band **194**. In these examples, the second outer recess **176** serves to retain the second retainer **168** around the second section **128**. In some examples, the second outer recess **176** represents a reduction in the outer dimension **144** (e.g., outer diameter) of a portion of the second section **128** relative to another portion of the second section **128**.

[0081] Referring to FIGS. **1** and **11**, in one or more examples, the insulated sleeve connector **100** includes the first inner recess **196**. The first inner recess **196** is disposed on (e.g., depends radially outward from) the inner surface **136** of the first section **126** of the tubular body **102**, for example, proximate (e.g., at or near) the first end **112**. In one or more examples of the insulated sleeve connector **100**, the first inner band **186** is disposed in the first inner recess **196**. In some examples, the first inner recess **196** represents an increase in the inner dimension **142** (e.g., inner diameter) of a portion of the first section **126** relative to another portion of the first section **126**.

[0082] Referring to FIGS. **1** and **11**, in one or more examples, the insulated sleeve connector **100** includes the second inner recess **198**. The second inner recess **198** is disposed on (e.g., depends radially outward from) the inner surface **136** of the second section **128** of the tubular body **102**, for example, proximate (e.g., at or near) the second end **116**. In one or more examples of the insulated sleeve connector **100**, the second inner band **188** is disposed in the second inner recess **198**. In some examples, the second inner recess **198** represents an increase in the inner dimension **142** (e.g., inner diameter) of a portion of the second section **128** relative to another portion of the second section **128**.

[0083] Referring to FIGS. **1** and **11-13**, in one or more examples of the insulated sleeve connector **100**, at least one, or each one, of the first inner band **186**, the first outer band **192**, the second inner band **188**, and/or the second outer band **194** includes relief cuts **178**. The relief cuts **178** enable expansion of the first inner band **186**, the first outer band **192**, the second inner band **188**, and/or the second outer band **194** and, thus, expansion of the tubular body **102** during insertion of one of the ducts **252**. FIG. **12** illustrates an example of the relief cuts **178** formed in a portion of one of the bands (e.g., first inner band **186**, first outer band **192**, second inner band **188**, and/or second outer band **194**). FIG. **13** illustrates an example of the portion of the band shown in FIG. **12** (e.g., first inner band **186**, first outer band **192**, second inner band **188**, and/or second outer band **194**) in an expanded state, thereby, enabling radial expansion of a portion of the tubular body **102** that includes the band (e.g., the first section **126** and/or the second section **128**).
[0084] Referring to FIG. **1**, in one or more examples of the insulated sleeve connector **100**, at least one, or each one, of the first inner band **186**, the first outer band **192**, the second inner band **188**, and/or the second outer band **194** includes or is made of tape **212**. In one or more examples, the tape **212** is an adhesive-backed strip tape.
[0085] Referring to FIGS. **1** and **11**, in one or more examples, the insulated sleeve connector **100** includes the first tabs **152**. The first tabs **152** are coupled to at least one of the first inner band **186** and first outer band **192**. The first tabs **152** extend outwardly from the first end **112** along the longitudinal axis **104**. In one or more examples, the first tabs **152** include or are formed by the tape **212**.
[0086] Referring to FIGS. **1** and **11**, in one or more examples, the insulated sleeve connector **100** includes a number of the second tabs **154**. The second tabs **154** are coupled to at least one of the second inner band **188** and the second outer band **194**. The second tabs **154** extend outwardly from the second end **116** along the longitudinal axis **104**. In one or more examples, the second tabs **154** include or are formed by the tape **212**.
[0087] Referring to FIGS. **1**, **14** and **15**, in one or more examples of the insulated sleeve connector **100**, the tubular body **102** includes the base layer **106**. The base layer **106** includes or is made of a flexible material **202**. In one or more examples, the flexible material **202** includes or is made of silicone. In one or more examples, the flexible material **202** includes or is made of a fiber-reinforced (e.g., fiberglass) substrate. The tubular body **102** also includes the insulated layer **108**. The insulated layer **108** is coupled to the base layer **106**. The insulated layer **108** includes or is made of the insulating material **124**. In one or more examples, the insulated layer **108** and the base layer **106** are bonded together.
[0088] Referring still to FIGS. **1**, **14** and **15**, in one or more examples of the insulated sleeve connector **100**, at least a portion of the insulated layer **108** is coupled to an inner side **208** of the base layer **106**, for example, facing the hollow interior of the tubular body **102** and the ducts **252**. In one or more examples of the insulated sleeve connector **100**, at least a portion of the insulated layer **108** is coupled to an outer side **206** of the base layer **106**, for example, facing away from the hollow interior of the tubular body **102** and the ducts **252**. In one or more examples, a portion of the insulated layer **108** is coupled to the outer side **206** of the base layer **106** and another portion of the insulated layer **108** is coupled to the inner side **208** of the base layer **106**. In various examples, the insulated layer **108** can be on the outer side **206**, the inner side **208**, or both along the central section **122** of the tubular body **102**. In other examples, the insulated layer **108** can be on the outer side **206**, the inner side **208**, or both along the first section **126** and/or the second section **128** of the tubular body **102**.
[0089] In one or more examples, the insulated layer **108** includes or is made up of a number of layers of the insulating material **124**. In one or more examples, a portion of the insulated layer **108** that is located on the inside of the insulated sleeve connector **100** (e.g., coupled to the inner side **208** of the base layer **106**) also provides a substantially similar effect as the first protrusion **160** and the second protrusion **170**, which is to provide positive tactile indication of engagement over the

ends and/or beads of the ducts **252**. In addition, in one or more examples, bands or strips of the insulated layer **108** can be used on the inside of the insulated sleeve connector **100** that include or are made of a closed cell foam or an elastomer to minimize air leakage.

[0090] Referring to FIGS. **14** and **15**, in one or more examples of the insulated sleeve connector **100**, the base layer **106** (e.g., the flexible material **202**) extends outwardly from the insulated layer **108** along the longitudinal axis **104** from the first end **112** to form the first tabs **152** that extend outwardly from the first end **112** and/or around the perimeter of the first opening **114**. Similarly, in one or more examples of the insulated sleeve connector **100**, the base layer **106** extends outwardly from the insulated layer **108** along the longitudinal axis **104** from the second end **116** to form the second tabs **154** the extend outwardly from the second end **116** and/or around the perimeter of the first opening **114**.

[0091] Referring now to FIGS. **1** and **3-15**, the following are examples of the duct system **250**, according to the present disclosure. The duct system **250** includes a number of elements, features, and components. Not all of the elements, features, and/or components described or illustrated in one example are required in that example. Some or all of the elements, features, and/or components described or illustrated in one example can be combined with other examples in various ways without the need to include other elements, features, and/or components described in those other examples, even though such combination or combinations are not explicitly described or illustrated by example herein. The examples of the duct system **250** can be utilized with any one of the examples of the insulated sleeve connector **100** described herein and illustrated in FIGS. **1** and **3-15**.

[0092] Referring to FIGS. **1** and **3-15**, in one or more examples, the duct system **250** includes the first duct **254**, the second duct **256**, and the insulated sleeve connector **100** connecting the first duct **254** and the second duct **256** together. The insulated sleeve connector **100** includes the tubular body **102** having the longitudinal axis **104**. The tubular body **102** includes the first end **112** that includes the first opening **114**. The tubular body **102** includes the second end **116**, opposite the first end **112** along the longitudinal axis **104**. that includes the second opening **118**. The tubular body **102** includes the central section **122** that is situated along the longitudinal axis **104** between the first end **112** and the second end **116**. The central section **122** includes or is made of the insulating material **124**.

[0093] Referring to FIGS. **1**, **3-10** **14** and **15**, in one or more examples of the duct system **250**, the insulated sleeve connector **100** includes the first protrusion **160**. The first protrusion **160** projects from the inner surface **136** of the tubular body **102** adjacent to the central section **122**. The insulated sleeve connector **100** includes the second protrusion **170**. The second protrusion **170** projects from the inner surface **136** of the tubular body **102** adjacent to the central section **122**. The second protrusion **170** is spaced away from the first protrusion **160** along the longitudinal axis **104**.

[0094] Referring to FIGS. **1**, **7** and **8**, in one or more examples of the duct system **250**, the insulated sleeve connector **100** includes the hinge **164** that is disposed on the tubular body **102** between the central section **122** and the second end **116**. A portion of the tubular body **102** is configured to fold back along the longitudinal axis **104** about the hinge **164**.

[0095] Referring generally to FIGS. **1** and **3-15** and particularly to FIG. **2**, the following are examples of the method **1000**, according to the present disclosure. In one or more examples, the method **1000** is implemented using the insulated sleeve connector **100** (FIGS. **1** and **3-15**). The method **1000** includes a number of elements, steps, operations, or processes. Not all of the elements, steps, operations, or processes described or illustrated in one example are required in that example. Some or all of the elements, steps, operations, or processes described or illustrated in one example can be combined with other examples in various ways without the need to include other elements, steps, operations, or processes described in those other examples, even though such combination or combinations are not explicitly described or illustrated by example herein.

[0096] In one or more examples, the method **1000** includes a step of inserting **1002** the first duct

**254** into the first end **112** of the tubular body **102** of the insulated sleeve connector **100**.

[0097] In one or more examples, the method **1000** includes a step of moving **1004** the first duct end **272** past the first protrusion **160**. The first protrusion **160** projects from the inner surface **136** of the tubular body **102** adjacent to the central section **122** of the tubular body **102**.

[0098] In one or more examples, the method **1000** includes a step of positioning **1006** the first duct end **272** of the first duct **254** within the central section **122** of the tubular body **102**.

[0099] In one or more examples, the method **1000** includes a step of positioning **1008** the first bead **262** in the first groove **132**. The first groove **132** is disposed on the inner surface **136** of the tubular body **102** along the central section **122**.

[0100] In one or more examples, the method **1000** includes a step of inserting **1010** the second duct **256** into the second end **116** of the tubular body **102**.

[0101] In one or more examples, the method **1000** includes a step of moving **1012** the second duct end **274** past the second protrusion **170**. The second protrusion **170** projects from the inner surface **136** of the tubular body **102** adjacent to the central section **122** and is spaced away from the first protrusion **160** along the longitudinal axis **104**.

[0102] In one or more examples, the method **1000** includes a step of positioning **1014** the second duct end **274** of the second duct **256** within the central section **122** of the tubular body **102**. In one or more examples, the first duct **254** and the second duct **256** are positioned such that the distance **286** between the first duct end **272** and the second duct end **274** is less than the length dimension **162** of the central section along the longitudinal axis **104**.

[0103] In one or more examples, the method **1000** includes a step of positioning **1016** the second bead **264** in the second groove **134**. The second groove **134** is disposed on the inner surface **136** of the tubular body **102** along the central section **122** and is spaced from the first groove **132** along the longitudinal axis **104**.

[0104] In one or more examples, the method **1000** includes a step of insulating **1018** the gap **258** between the first duct end **272** and the second duct end **274** with the central section **122**. In one or more examples, the step of insulating **1018** is performed or achieved by the insulated sleeve connector **100** being connected to the end portions of the ducts **252**, for example, by the insulating material **124** used to form at least a portion of the insulated sleeve connector **100**, the thickness of the tubular body **102**, the thickness of the insulating material **124**, the relative position of the insulating material **124** and/or the insulated layer **108**, and the like.

[0105] In one or more examples, the method **1000** includes a step of positioning **1020** the first retainer **166** along the first section **126** of the tubular body **102** extending along the longitudinal axis **104** between the first end **112** and the central section **122**.

[0106] In one or more examples, the method **1000** includes a step of positioning **1022** the second retainer **168** along the second section **128** of the tubular body **102** extending along the longitudinal axis **104** between the second end **116** and the central section **122**.

[0107] In one or more examples, the method **1000** includes a step of folding **1024** the second section **128** of the tubular body **102**, extending along the longitudinal axis **104** between the second end **116** and the central section **122**, back along the longitudinal axis **104** about the hinge **164**. The step of folding **1024** reduces the length of the tubular body **102** for installation on one of the ducts **252**. In one or more examples, the step of folding **1024** is performed before the step of inserting **1010** the second duct **256** into the insulated sleeve connector **100** (e.g., FIG. **7**).

[0108] In one or more examples, the method **1000** includes a step of unfolding **1026** the second section **128** of the tubular body **102** along the longitudinal axis **104** about the hinge **164**. The step of unfolding **1026** returns the tubular body **102** to its original length for installation on one of the ducts **252**. In one or more examples, the step of unfolding **1026** is performed after the step of inserting **1010** the second duct **256** into the insulated sleeve connector **100**, such as after the step of positioning **1014** the second duct **256** relative to the central section **122** (FIG. **8**).

[0109] In one or more examples, according to the method **1000**, the step of positioning **1022** the second retainer **168** includes a step of moving the second retainer **168** along the ramp **172** between the first section **126** and the second section **128** to the hinge **164**.

[0110] Referring now to FIGS. **16** and **17**, examples of the insulated sleeve connector **100**, the duct system **250**, and the method **1000** described herein, may be related to, or used in the context of, the aerospace manufacturing and service method **1100**, as shown in the flow diagram of FIG. **16** and an aircraft **1200**, as schematically illustrated in FIG. **17**. As an example, the aircraft **1200** and/or the manufacturing and service method **1100** may include or utilize the duct system **250** that includes the ducts **252** connected using the insulated sleeve connector **100**.

[0111] Referring to FIG. **17**, which illustrates an example of the aircraft **1200**. The aircraft **1200** can be any aerospace vehicle or platform. In one or more examples, the aircraft **1200** includes the airframe **1202** having the interior **1206**. The aircraft **1200** includes a plurality of onboard systems **1204** (e.g., high-level systems). Examples of the onboard systems **1204** of the aircraft **1200** include propulsion systems **1208**, hydraulic systems **1212**, electrical systems **1210**, and environmental systems **1214**. In other examples, the onboard systems **1204** also includes one or more control systems coupled to the airframe **1202** of the aircraft **1200**. In yet other examples, the onboard systems **1204** also include one or more other systems **1216**, such as, but not limited to, communications systems, avionics systems, software distribution systems, network communications systems, passenger information/entertainment systems, guidance systems, radar systems, weapons systems, and the like. The aircraft **1200** can include the duct system **250** having the ducts **252** that are connected together using the insulated sleeve connector **100**.

[0112] Referring to FIG. **16**, during pre-production of the aircraft **1200**, the manufacturing and service method **1100** includes specification and design of the aircraft **1200** (block **1102**) and material procurement (block **1104**). During production of the aircraft **1200**, component and subassembly manufacturing (block **1106**) and system integration (block **1108**) of the aircraft **1200** take place. Thereafter, the aircraft **1200** goes through certification and delivery (block **1110**) to be placed in service (block **1112**). Routine maintenance and service (block **1114**) includes modification, reconfiguration, refurbishment, etc. of one or more systems of the aircraft **1200**.

[0113] Each of the processes of the manufacturing and service method **1100** illustrated in FIG. **16** may be performed or carried out by a system integrator, a third party, and/or an operator (e.g., a customer). For the purposes of this description, a system integrator may include, without limitation, any number of aircraft manufacturers and major-system subcontractors; a third party may include, without limitation, any number of vendors, subcontractors, and suppliers; and an operator may be an airline, leasing company, military entity, service organization, and so on.

[0114] Examples of the insulated sleeve connector **100**, the duct system **250**, and the method **1000**, shown and described herein, may be employed during any one or more of the stages of the manufacturing and service method **1100** shown in the flow diagram illustrated by FIG. **16**. In an example, the duct system **250** of the aircraft **1200** can be installed or connected using the insulated sleeve connector **100** and/or according to the method **1000** during a portion of component and subassembly manufacturing (block **1106**) and/or system integration (block **1108**). Further, the duct system **250** of the aircraft **1200** can be installed or connected using the insulated sleeve connector **100** and/or according to the method **1000** while the aircraft **1200** is in service (block **1112**). Also, the duct system **250** of the aircraft **1200** can be installed or connected using the insulated sleeve connector **100** and/or according to the method **1000** during system integration (block **1108**) and certification and delivery (block **1110**). Similarly, the duct system **250** of the aircraft **1200** can be installed or connected using the insulated sleeve connector **100** and/or according to the method **1000** while the aircraft **1200** is in service (block **1112**) and during maintenance and service (block **1114**).

[0115] The preceding detailed description refers to the accompanying drawings, which illustrate specific examples described by the present disclosure. Other examples having different structures and operations do not depart from the scope of the present disclosure. Like reference numerals may refer to the same feature, element, or component in the different drawings. Throughout the present disclosure, any one of a plurality of items may be referred to individually as the item and a plurality of items may be referred to collectively as the items and may be referred to with like reference numerals. Moreover, as used herein, a feature, element, component, or step preceded with the word "a" or "an" should be understood as not excluding a plurality of features, elements, components, or steps, unless such exclusion is explicitly recited.

[0116] Illustrative, non-exhaustive examples, which may be, but are not necessarily, claimed, of the subject matter according to the present disclosure are provided above. Reference herein to "example" means that one or more feature, structure, element, component, characteristic, and/or operational step described in connection with the example is included in at least one aspect, embodiment, and/or implementation of the subject matter according to the present disclosure. Thus, the phrases "an example," "another example," "one or more examples," and similar language throughout the present disclosure may, but do not necessarily, refer to the same example. Further, the subject matter characterizing any one example may, but does not necessarily, include the subject matter characterizing any other example. Moreover, the subject matter characterizing any one example may be, but is not necessarily, combined with the subject matter characterizing any other example.

[0117] As used herein, a system, apparatus, device, structure, article, element, component, or hardware "configured to" perform a specified function is indeed capable of performing the specified function without any alteration, rather than merely having potential to perform the specified function after further modification. In other words, the system, apparatus, device, structure, article, element, component, or hardware "configured to" perform a specified function is specifically selected, created, implemented, utilized, programmed, and/or designed for the purpose of performing the specified function. As used herein, "configured to" denotes existing characteristics of a system, apparatus, structure, article, element, component, or hardware that enable the system, apparatus, structure, article, element, component, or hardware to perform the specified function without further modification. For purposes of this disclosure, a system, apparatus, device, structure, article, element, component, or hardware described as being "configured to" perform a particular function may additionally or alternatively be described as being "adapted to" and/or as being "operative to" perform that function.

[0118] Unless otherwise indicated, the terms "first," "second," "third," etc. are used herein merely as labels, and are not intended to impose ordinal, positional, or hierarchical requirements on the items to which these terms refer. Moreover, reference to, e.g., a "second" item does not require or preclude the existence of, e.g., a "first" or lower-numbered item, and/or, e.g., a "third" or higher-numbered item.

[0119] As used herein, the phrase "at least one of," when used with a list of items, means different combinations of one or more of the listed items may be used and only one of each item in the list may be needed. For example, "at least one of item A, item B, and item C" may include, without limitation, item A or item A and item B. This example also may include item A, item B, and item C, or item B and item C. In other examples, "at least one of"' may be, for example, without limitation, two of item A, one of item B, and ten of item C; four of item B and seven of item C; and other suitable combinations. As used herein, the term "and/or" and the "/" symbol includes any and all combinations of one or more of the associated listed items.

[0120] For the purpose of this disclosure, the terms "coupled," "coupling," and similar terms refer to two or more elements that are joined, linked, fastened, attached, connected, put in communication, or otherwise associated (e.g., mechanically, electrically, fluidly, optically, electromagnetically) with one another. In various examples, the elements may be associated directly or indirectly. As an example, clement A may be directly associated with element B. As another example, element A may be indirectly associated with element B, for example, via another element C. It will be understood that not all associations among the various disclosed elements are

necessarily represented. Accordingly, couplings other than those depicted in the figures may also exist.
[0121] As used herein, the term “approximately” refers to or represents a condition that is close to, but not exactly, the stated condition that still performs the desired function or achieves the desired result. As an example, the term “approximately” refers to a condition that is within an acceptable predetermined tolerance or accuracy, such as to a condition that is within 10% of the stated condition. However, the term “approximately” does not exclude a condition that is exactly the stated condition. As used herein, the term “substantially” refers to a condition that is essentially the stated condition that performs the desired function or achieves the desired result.
[0122] FIGS. **1**, **3-5** and **17**, referred to above, may represent functional elements, features, or components thereof and do not necessarily imply any particular structure. Accordingly, modifications, additions and/or omissions may be made to the illustrated structure. Additionally, those skilled in the art will appreciate that not all elements, features, and/or components described and illustrated in FIGS. **1**, **3-5** and **17**, referred to above, need be included in every example and not all elements, features, and/or components described herein are necessarily depicted in each illustrative example. Accordingly, some of the elements, features, and/or components described and illustrated in FIGS. **1**, **3-5** and **17** may be combined in various ways without the need to include other features described and illustrated in FIGS. **1**, **3-5** and **17**, other drawing figures, and/or the accompanying disclosure, even though such combination or combinations are not explicitly illustrated herein. Similarly, additional features not limited to the examples presented, may be combined with some or all of the features shown and described herein. Unless otherwise explicitly stated, the schematic illustrations of the examples depicted in FIGS. **1**, **3-5** and **17**, referred to above, are not meant to imply structural limitations with respect to the illustrative example. Rather, although one illustrative structure is indicated, it is to be understood that the structure may be modified when appropriate. Accordingly, modifications, additions and/or omissions may be made to the illustrated structure. Furthermore, elements, features, and/or components that serve a similar, or at least substantially similar, purpose are labeled with like numbers in each of FIGS. **1**, **3-5** and **17**, and such elements, features, and/or components may not be discussed in detail herein with reference to each of FIGS. **1**, **3-5** and **17**. Similarly, all elements, features, and/or components may not be labeled in each of FIGS. **1**, **3-5** and **17**, but reference numerals associated therewith may be utilized herein for consistency.
[0123] In FIGS. **2** and **16**, referred to above, the blocks may represent operations, steps, and/or portions thereof and lines connecting the various blocks do not imply any particular order or dependency of the operations or portions thereof. It will be understood that not all dependencies among the various disclosed operations are necessarily represented. FIGS. **2** and **16** and the accompanying disclosure describing the operations of the disclosed methods set forth herein should not be interpreted as necessarily determining a sequence in which the operations are to be performed. Rather, although one illustrative order is indicated, it is to be understood that the sequence of the operations may be modified when appropriate. Accordingly, modifications, additions and/or omissions may be made to the operations illustrated and certain operations may be performed in a different order or simultaneously. Additionally, those skilled in the art will appreciate that not all operations described need be performed.
[0124] Further, references throughout the present specification to features, advantages, or similar language used herein do not imply that all of the features and advantages that may be realized with the examples disclosed herein should be, or are in, any single example. Rather, language referring to the features and advantages is understood to mean that a specific feature, advantage, or characteristic described in connection with an example is included in at least one example. Thus, discussion of features, advantages, and similar language used throughout the present disclosure may, but does not necessarily, refer to the same example.
[0125] The described features, advantages, and characteristics of one example may be combined in

any suitable manner in one or more other examples. One skilled in the relevant art will recognize that the examples described herein may be practiced without one or more of the specific features or advantages of a particular example. In other instances, additional features and advantages may be recognized in certain examples that may not be present in all examples. Furthermore, although various examples of the insulated sleeve connector **100**, the duct system **250**, and the method **1000** have been shown and described, modifications may occur to those skilled in the art upon reading the specification. The present application includes such modifications and is limited only by the scope of the claims.

## Claims

**1**. An insulated sleeve connector comprising: a tubular body having a longitudinal axis and comprising: a first end comprising a first opening; a second end opposite the first end along the longitudinal axis and comprising a second opening; and a central section situated along the longitudinal axis between the first end and the second end and comprising an insulating material; a first protrusion projecting from an inner surface of the tubular body adjacent to the central section; and a second protrusion projecting from the inner surface of the tubular body adjacent to the central section and spaced away from the first protrusion along the longitudinal axis.
**2-3**. (canceled)
**4**. The insulated sleeve connector of claim 1, wherein: the tubular body further comprises: a first section extending along the longitudinal axis between the first end and the central section; and a second section extending along the longitudinal axis between the second end and the central section; the first section of the tubular body is configured to receive a first duct; the second section of the tubular body is configured to receive a second duct; and the central section of the tubular body has a length dimension along the longitudinal axis that is greater than a distance between a first duct end of the first duct and a second duct end of the second duct when the first duct is received by the first section and the second duct is received by the second section.
**5**. The insulated sleeve connector of claim 4, further comprising: a first groove disposed on an inner surface of the tubular body along the central section, wherein the first groove is configured to receive a first bead of the first duct; and a second groove disposed on the inner surface of the tubular body along the central section and spaced from the first groove along the longitudinal axis, wherein the second groove is configured to receive a second bead of the second duct.
**6**. The insulated sleeve connector of claim 5, wherein: the first groove has a first groove width that is larger than a first bead width of the first bead; and the second groove has a second groove width that is larger than a second bead width of the second bead.
**7**. The insulated sleeve connector of claim 4, wherein at least one of the first section and the second section of the tubular body comprises the insulating material.
**8**. The insulated sleeve connector of claim 4, wherein: the first protrusion is located at a first junction of the first section and the central section and provides a first tactile response to insertion of the first duct; and the second protrusion is located at a second junction of the second section and the central section that provides a second tactile response to insertion of the second duct.
**9**. The insulated sleeve connector of claim 4, wherein: the second section of the tubular body comprises a hinge; and a portion of the second section is configured to fold back along the longitudinal axis about the hinge.
**10**. The insulated sleeve connector of claim 9, further comprising a second retainer, wherein the hinge is configured to receive the second retainer.
**11**. The insulated sleeve connector of claim 10, wherein: the tubular body has an outer dimension along the first section that increases from the first end to the central section to form a ramp; and the second retainer is movable along the ramp between the first section and the second section.
**12**. The insulated sleeve connector of claim 1, further comprising: a number of first tabs extending

outwardly from the first end along the longitudinal axis; and a number of second tabs extending outwardly from the second end along the longitudinal axis.

**13-14**. (canceled)

**15**. The insulated sleeve connector of claim 1, further comprising: a first flange extending from the tubular body at the first end; and a second flange extending from the tubular body at the second end.

**16**. The insulated sleeve connector of claim 1, further comprising: a first inner band coupled to an inner surface of the tubular body at the first end; and a second inner band coupled to the inner surface of the tubular body at the second end.

**17**. The insulated sleeve connector of claim 16, further comprising: a first outer band coupled to an outer surface of the tubular body at the first end; and a second outer band coupled to the outer surface of the tubular body at the second end.

**18-20**. (canceled)

**21**. The insulated sleeve connector of claim 1, further comprising: a number of first tabs coupled to at least one of the first inner band and first outer band and extending outwardly from the first end along the longitudinal axis; and a number of second tabs coupled to at least one of the second inner band and the second outer band and extending outwardly from the second end along the longitudinal axis.

**22**. The insulated sleeve connector of claim 1, wherein the tubular body comprises: a base layer comprising a flexible material; and an insulated layer coupled to the base layer and comprising the insulating material.

**23-24**. (canceled)

**25**. The insulated sleeve connector of claim **2**, wherein the base layer extends outwardly from the insulated layer along the longitudinal axis to form a number of first tabs extending outwardly from the first end and a number of second tabs extending outwardly from the second end.

**26-50**. (canceled)

**51**. An insulated sleeve connector comprising: a tubular body having a longitudinal axis and comprising: a first end comprising a first opening; a second end opposite the first end along the longitudinal axis and comprising a second opening; and a central section situated along the longitudinal axis between the first end and the second end and comprising an insulating material; and a hinge disposed on the tubular body between the central section and the second end, wherein a portion of the tubular body is configured to fold back along the longitudinal axis about the hinge.

**52**. The insulated sleeve connector of claim 51, wherein: the tubular body further comprises: a first section extending along the longitudinal axis between the first end and the central section; and a second section extending along the longitudinal axis between the second end and the central section; the first section of the tubular body is configured to receive a first duct; the second section of the tubular body is configured to receive a second duct; and the central section of the tubular body has a length dimension along the longitudinal axis that is greater than a distance between a first duct end of the first duct and a second duct end of the second duct when the first duct is received by the first section and the second duct is received by the second section.

**53-77**. (canceled)

**78**. A method for connecting ducts using an insulated sleeve connector, the method comprising: inserting a first duct into a first end of a tubular body of the insulated sleeve connector; moving the first duct end past a first protrusion projecting from an inner surface of the tubular body adjacent to a central section of the tubular body situated along a longitudinal axis of the tubular body between the first end and a second end of the tubular body; positioning a first duct end of the first duct within the central section of the tubular body; inserting a second duct into a second end of the tubular body; moving a second duct end past a second protrusion projecting from the inner surface of the tubular body adjacent to the central section and spaced away from the first protrusion along the longitudinal axis; positioning the second duct end of the second duct within the central section of the tubular body such that a distance between the first duct end and the second duct end is less than a length dimension of the central section along the longitudinal axis; and insulating a gap between the first duct end and the second duct end with the central section.

**79-80**. (canceled)

**81**. The method of claim 78, further comprising folding a second section of the tubular body, extending along the longitudinal axis between the second end and the central section, back along the longitudinal axis about a hinge before inserting the second duct into the second end of the tubular body.

**82-87**. (canceled)